OXFORD AQA
INTERNATIONAL QUALIFICATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname ____________________

Forename(s) ____________________

Candidate signature ____________________
I declare this is my own work.

INTERNATIONAL GCSE
CHEMISTRY

Paper 2

Tuesday 27 May 2025 07:00 GMT Time allowed: 1 hour 30 minutes

Materials
For this paper you must have:
- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).

Instructions
- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	

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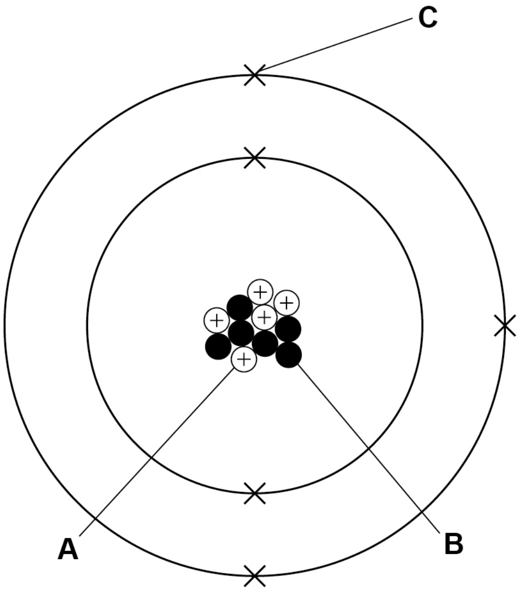
Answer **all** questions in the spaces provided.

0 1 This question is about boron and carbon.

Figure 1 represents a boron atom.

Figure 1



0 1 . 1 Atoms are made of electrons, neutrons and protons.

Draw **one** line from each letter to the sub-atomic particle the letter represents.

Use **Figure 1**.

[2 marks]

Letter	**Sub-atomic particle**
A	Electron
B	Neutron
C	Proton

Do not write outside the box

0 1 . 2 Which group of the periodic table is boron in?

Use **Figure 1**.

[1 mark]

Tick (✓) **one** box.

Group 2 ☐

Group 3 ☐

Group 4 ☐

Group 5 ☐

0 1 . 3 Boron is an element.

What is meant by 'an element'?

[1 mark]

____

____

Question 1 continues on the next page

Do not write outside the box

Figure 2 represents two different types of carbon atom.

Figure 2

$^{12}_{6}C$ $^{14}_{6}C$

0 1 . 4 What are different types of atom of the same element called?

[1 mark]

Tick (✓) **one** box.

Alloys ☐

Isotopes ☐

Molecules ☐

0 1 . 5 **Table 1** shows the numbers of sub-atomic particles present in two different atoms of carbon.

Table 1

Sub-atomic particle	$^{12}_{6}C$	$^{14}_{6}C$
Electron	6	
Neutron	6	
Proton		6

Complete **Table 1**.

[3 marks]

8

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Turn over for the next question

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ANSWER IN THE SPACES PROVIDED

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0 2 This question is about polymers.

Hydrogels are polymers that absorb water.

A student investigated the change in mass of three different types of hydrogel when placed in water.

This is the method used.

1 Add 1.2 g of hydrogel to a beaker.
2 Add 100 cm^3 of distilled water to the beaker.
3 Leave the hydrogel and water for 10 minutes.
4 Filter the mixture.
5 Remove the hydrogel from the filter paper.
6 Weigh the hydrogel again.
7 Repeat steps **1** to **6** with a different type of hydrogel.

0 2 . 1 Identify the variables used in this investigation.

Draw **one** line from each variable to the description of the variable.

[2 marks]

Variable	**Description of variable**
	Change in mass of hydrogel
	Initial mass of hydrogel
Dependent	
	Time hydrogel left in water
Independent	
	Type of hydrogel
	Volume of distilled water

Do not write outside the box

0 2 . 2 The student observed that some hydrogel remained stuck to the filter paper.

What effect would this have on the measured change in mass?

[1 mark]

Tick (✓) **one** box.

Measured change in mass would be lower. ☐

Measured change in mass would be the same. ☐

Measured change in mass would be higher. ☐

0 2 . 3 **Table 2** shows the results for one type of hydrogel.

Table 2

Initial mass of hydrogel in grams	Final mass of hydrogel in grams
1.3	66.3

Calculate the ratio of the mass of water the hydrogel absorbed to the initial mass of the hydrogel.

[3 marks]

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Mass of water the hydrogel absorbed **:** initial mass of hydrogel = ____________ **:** 1

Question 2 continues on the next page

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0 2 . 4 Polymers have other uses.

Which is also a use of polymers?

[1 mark]

Tick (✓) **one** box.

Bleach ☐

Packaging ☐

Solvents ☐

0 2 . 5 Polymers can be either thermosetting or thermosoftening.

Explain why thermosetting polymers do **not** change shape when heated.

[2 marks]

9

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0 3 Alkanes are hydrocarbons.

0 3 . 1 An alkane has the formula C_3H_8

What is the name of this alkane?

[1 mark]

Tick (✓) **one** box.

Ethane ☐

Methane ☐

Propane ☐

0 3 . 2 What is the general formula for alkanes?

[1 mark]

Tick (✓) **one** box.

C_nH_n ☐

C_nH_{n+2} ☐

C_nH_{2n} ☐

C_nH_{2n+2} ☐

0 3 . 3 Alkanes are saturated hydrocarbons.

What is meant by 'saturated'?

[1 mark]

Tick (✓) **one** box.

All the carbon-carbon bonds are double covalent bonds. ☐

All the carbon-carbon bonds are single covalent bonds. ☐

The carbon-carbon bonds are a mixture of single and double covalent bonds. ☐

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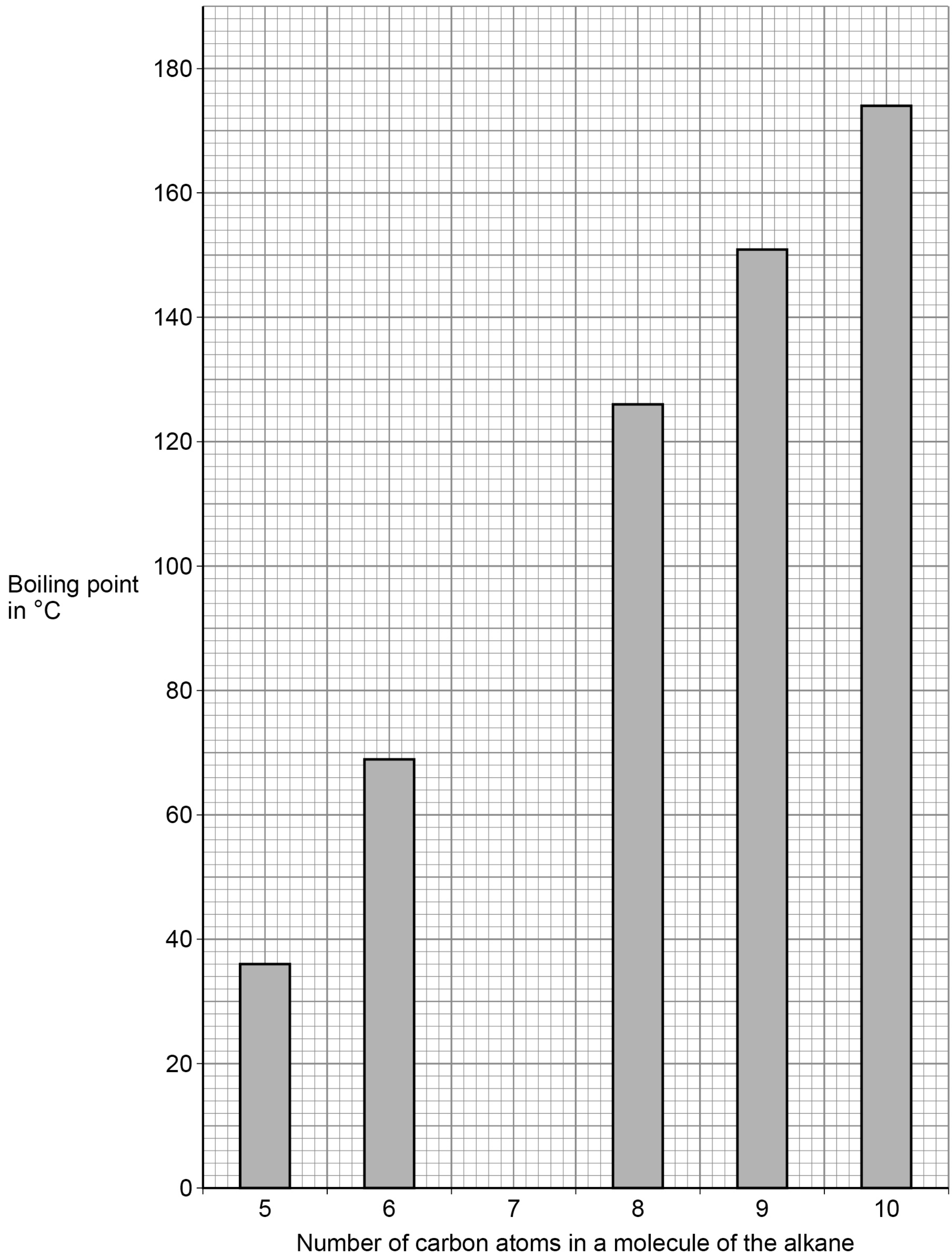
Figure 3 shows the boiling points of some alkanes.

Figure 3



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0 3 . 4 Predict the boiling point of the alkane with 7 carbon atoms in a molecule.

Use **Figure 3**.

[1 mark]

________________ °C

0 3 . 5 Complete the sentence to describe the trend in **Figure 3**.

[1 mark]

As the number of carbon atoms in a molecule increases, the boiling point

__________________________________ .

0 3 . 6 Crude oil is a mixture of alkanes.

Crude oil is separated by fractional distillation.

Alkane **X** condenses at a lower position in a fractionating column than the alkanes in **Figure 3**.

Explain why.

[3 marks]

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Turn over for the next question

Do not write outside the box

0 4 Vehicles can be powered by:

- a hydrogen combustion engine
- a hydrogen fuel cell.

Table 3 gives information about hydrogen combustion engines and hydrogen fuel cells.

Table 3

	Hydrogen combustion engine	**Hydrogen fuel cell**
Maximum distance travelled before replacement in kilometres	300 000	190 000
Distance travelled before refuelling in kilometres	200	500

0 4 . 1 Explain **one** advantage of a hydrogen fuel cell.

Use **Table 3**.

[2 marks]

0 4 . 2 Explain **one** advantage of a hydrogen combustion engine.

Use **Table 3**.

[2 marks]

Do not write outside the box

0 4 . 3 Give **one** disadvantage of using hydrogen as fuel.

Do **not** use information from **Table 3**.

[1 mark]

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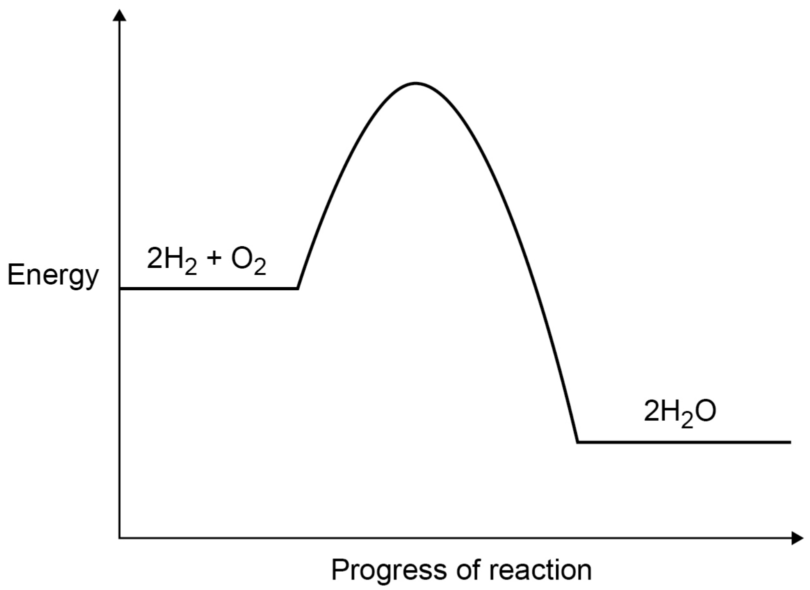
Figure 4 shows the energy level diagram for the combustion of hydrogen.

Figure 4



0 4 . 4 Hydrogen combustion engines use a catalyst.

Complete the energy level diagram in **Figure 4** to show the effect of using a catalyst.

You should:

- draw the line to show the progress of the reaction using a catalyst
- label the activation energy of the reaction using a catalyst.

[2 marks]

0 4 . 5 How does the energy level diagram show that the reaction is exothermic?

Use **Figure 4**.

[1 mark]

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Turn over ►

Do not write outside the box

0 4 . 6 Oxides of nitrogen can be produced in a hydrogen combustion engine.

Give **one** environmental problem caused by the release of oxides of nitrogen into the atmosphere.

[1 mark]

9

Do not write outside the box

0 5 Aluminium and copper are metals.

0 5 . 1 Aluminium is a pure metal.

Describe the structure of aluminium.

You should refer to the arrangement of aluminium atoms in your answer.

[2 marks]

0 5 . 2 Explain why aluminium is a good conductor of heat.

[2 marks]

Question 5 continues on the next page

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0 5 . 3 Overhead power cables are used to conduct electricity over large distances.

Figure 5 shows overhead power cables.

Figure 5

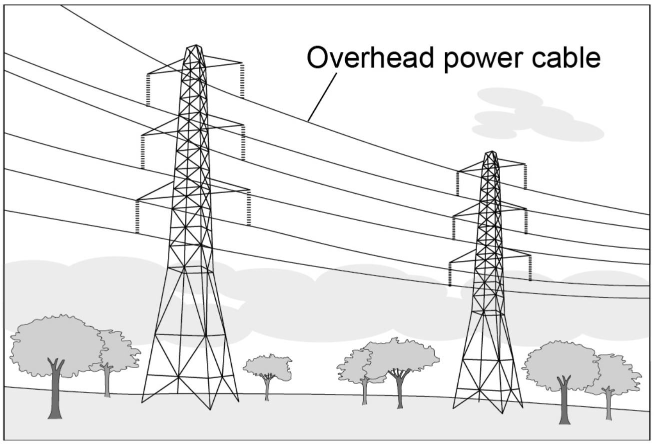


Table 4 gives information about aluminium and copper.

Table 4

	Aluminium	**Copper**
Density in g/cm^3	2.70	8.96
Melting point in °C	660	1085
Relative electrical conductivity	0.60	1.00
Relative cost per kilogram	1.00	3.76

Do not write outside the box

Evaluate the use of aluminium and of copper for overhead power cables.

[4 marks]

Extra space

8

Turn over for the next question

Do not write outside the box

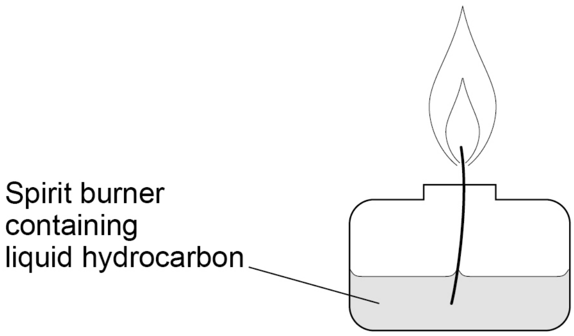
0 6 This question is about burning hydrocarbons.

0 6 . 1 A student did an experiment to determine the amount of energy released by burning a liquid hydrocarbon.

The student used a spirit burner to burn the liquid hydrocarbon.

Figure 6 shows a spirit burner.

Figure 6



Describe a method to determine the amount of energy released by burning 1 g of liquid hydrocarbon fuel.

[6 marks]

Do not write outside the box

Extra space

Table 5 shows the energy released by five different hydrocarbons when burnt.

Table 5

Formula of hydrocarbon	**Energy released in kJ/g**
C_5H_{12}	48.6
$C_{10}H_{22}$	46.0
$C_{15}H_{32}$	44.8
$C_{20}H_{42}$	47.2
$C_{25}H_{52}$	41.5

0 6 . 2 Describe the trend between the size of the hydrocarbon molecules and the energy released when burnt.

Use **Table 5**.

[2 marks]

0 6 . 3 When hydrocarbons are burnt, solid particulates can be produced.

Give the reason why solid particulates are produced.

[1 mark]

9

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0 7 Copper can be extracted from low-grade ores using phytomining.

Phytomining uses plants to absorb metal compounds.

0 7 . 1 Describe how copper compounds are obtained from plants.

[2 marks]

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0 7 . 2 Give **two** advantages of using phytomining instead of using traditional mining methods.

[2 marks]

1 ____

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2 ____

____

0 7 . 3 4650 g of copper compound was extracted using phytomining.

0.775 g of copper compounds can be extracted from one plant.

Calculate the number of plants used.

[2 marks]

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Number of plants = ____________________

Do not write outside the box

0 7 . 4 Suggest **two** disadvantages of phytomining.

[2 marks]

1 ____

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2 ____

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0 7 . 5 **Table 6** shows the energy required for mining and extracting copper from ore compared to recycling copper.

Table 6

Energy needed to mine and extract copper in kJ/kg	Energy needed to recycle copper in kJ/kg
8.25×10^7	1.24×10^7

Explain which process would have a lower carbon footprint.

[3 marks]

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11

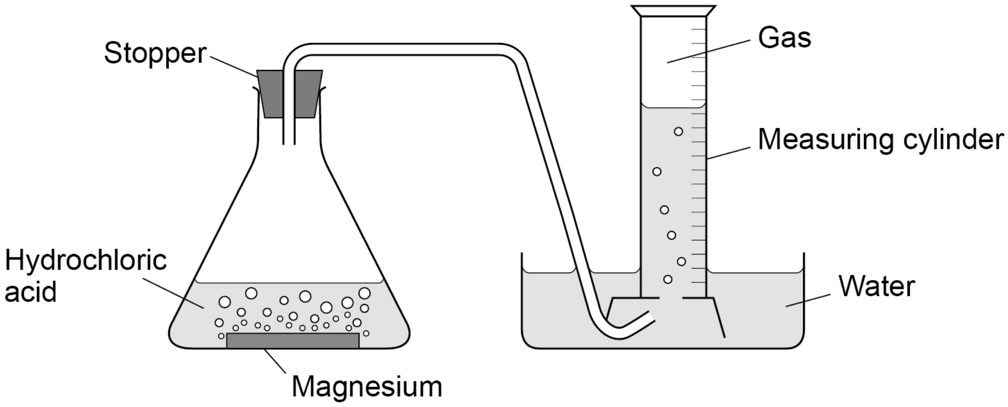
Turn over for the next question

Do not write outside the box

0 8 A student investigated the reaction of magnesium with hydrochloric acid.

Figure 7 shows the apparatus.

Figure 7



This is the method used:

1 Add 25 cm^3 of hydrochloric acid to the conical flask.
2 Add a 4 cm length of magnesium to the hydrochloric acid.
3 Insert the stopper into the conical flask as quickly as possible and start the timer.
4 Measure the volume of gas collected every 10 seconds until the reaction has finished.

0 8 . 1 Give **one** hazard associated with this investigation.

[1 mark]

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0 8 . 2 Give **one** observation that would show the reaction had finished.

[1 mark]

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Do not write outside the box

0 8 . 3 Why is the stopper inserted into the conical flask as quickly as possible in step **3**?

[1 mark]

Question 8 continues on the next page

Do not write outside the box

Table 7 shows the results for 1.0 mol/dm^3 hydrochloric acid.

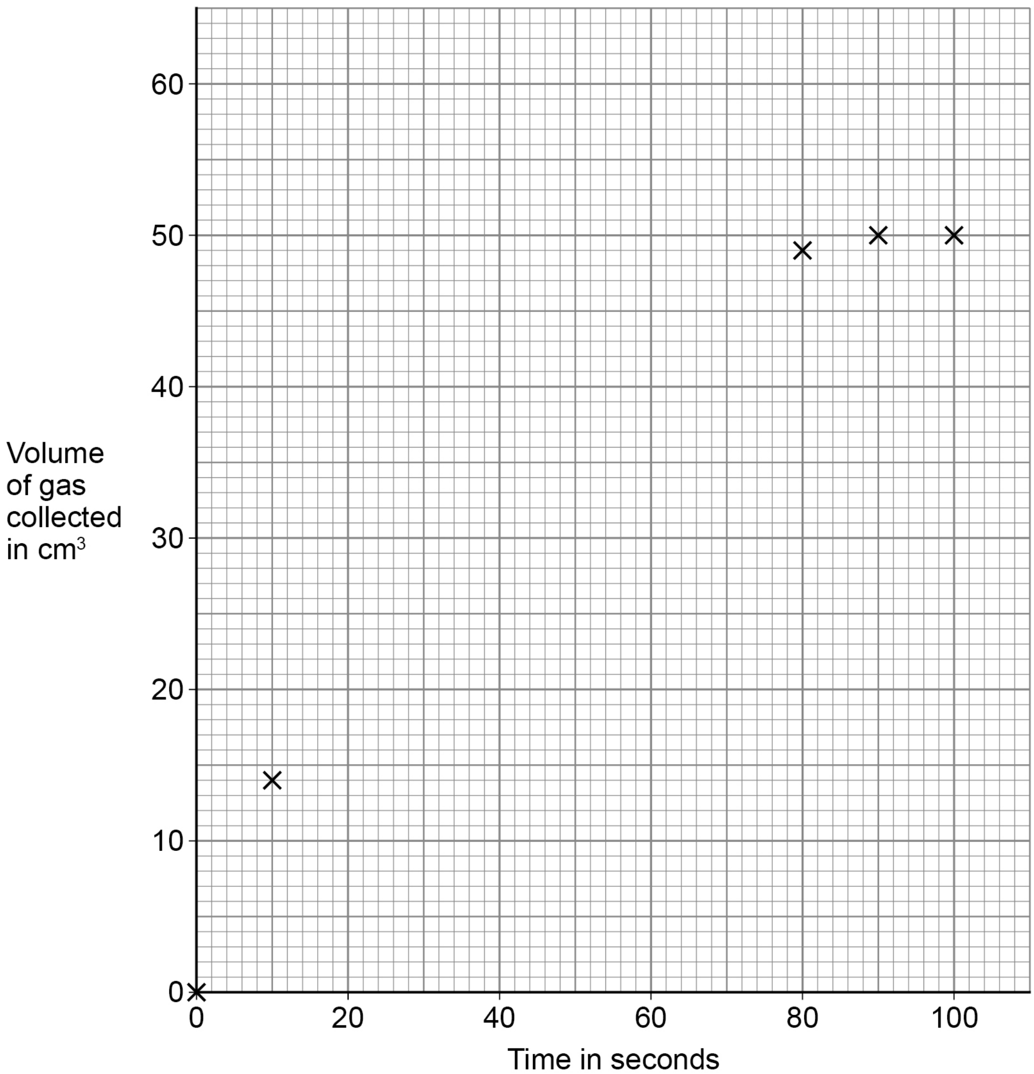
Table 7

Time in seconds	Volume of gas collected in cm^3
0	0
10	14
20	22
30	29
40	35
50	40
60	44
70	47
80	49
90	50
100	50

Do not write outside the box

0 8 . 4 **Figure 8** shows the volume of gas collected over time.

Figure 8



Complete **Figure 8**.

Some points have been plotted for you.

You should:

- plot the data from **Table 7**
- draw a line of best fit.

[3 marks]

0 8 . 5 Determine when the reaction using 1.0 mol/dm^3 hydrochloric acid finished.

Use **Figure 8**.

[1 mark]

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Do not write outside the box

Table 7 is repeated below.

Table 7

Time in seconds	Volume of gas collected in cm^3
0	0
10	14
20	22
30	29
40	35
50	40
60	44
70	47
80	49
90	50
100	50

0 8 . 6 Explain why the rate of reaction changes as the time increases.

Use **Table 7**.

[3 marks]

Do not write outside the box

0 8 . 7 Results can be described as reproducible or repeatable.

Give **two** ways in which reproducible results are different from repeatable results.

[2 marks]

1 ____

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2 ____

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0 8 . 8 Hydrochloric acid (HCl) ionises completely in water to produce hydrogen ions.

Calculate the number of hydrogen ions in 25 cm^3 of 2.0 mol/dm^3 hydrochloric acid.

One mole of hydrogen ions contains 6.02×10^{23} ions.

[4 marks]

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Number of hydrogen ions = ____________

16

Turn over for the next question

Do not write outside the box

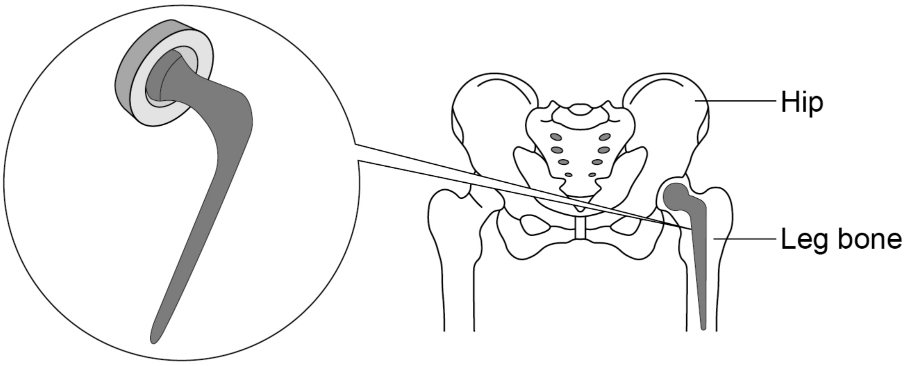
0 9 This question is about titanium and a titanium alloy.

Titanium is mixed with aluminium and vanadium to form an alloy.

The titanium alloy can be used to make artificial hip joints.

Figure 9 shows an artificial hip joint.

Figure 9



0 9 . 1 The titanium alloy used to make artificial hip joints is:

- strong
- resistant to corrosion.

Give **one** reason why each of these properties are needed in an artificial hip joint.

[2 marks]

Strength ____

____

Resistance to corrosion ____

____

Do not write outside the box

Titanium is extracted from an ore containing titanium dioxide in a two-stage process.

In **stage 1**, titanium dioxide is heated with carbon and chlorine to produce titanium chloride.

The equation for the reaction is:

$$TiO_2 + 2Cl_2 + 2C \rightarrow TiCl_4 + 2CO$$

0 9 . 2 720 g of titanium dioxide was completely reacted with chlorine and carbon.

Calculate the volume of chlorine gas used in dm^3.

Relative atomic masses (A_r) O = 16 Ti = 48

The volume of one mole of chlorine gas is 24.0 dm^3.

[5 marks]

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Volume of chlorine gas = _______________ dm^3

Question 9 continues on the next page

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In **stage 2**, titanium chloride ($TiCl_4$) reacts with sodium to produce titanium and sodium chloride.

0 9 . 3 Write the balanced symbol equation for the reaction to produce titanium from titanium chloride.

[2 marks]

__________ + __________ → __________ + __________

0 9 . 4 Water must be kept away from the reaction in **stage 2**.

Give **one** reason why it would be hazardous if water came into contact with sodium.

[1 mark]

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0 9 . 5 The reaction in **stage 2** is carried out in an atmosphere of argon and **not** in air.

Suggest why.

[2 marks]

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12

END OF QUESTIONS

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